LAB 14 - PROMPTS

Pairs Stage: a Hedged, Market-Neutral Statistical-Arbitrage Sleeve

Open your AI coding assistant (for example Claude Code or GitHub Copilot) in the labs_fullcode folder. Work down the steps: copy each prompt into the assistant, review the code it writes, then run the command. The complete, working code is already in Lab14-pairs-trading/solution/ if you want to compare. Never paste API keys into an AI chat.

Step 1: Read the method (K6)

Do this (no prompt needed)
Read the docstring of Lab14-pairs-trading/solution/pairs.py: the formation period 2018-2022 is used only to estimate the hedge ratio and test cointegration; the trading period 2023-2025 is out of sample with the hedge ratio held fixed.

Check: You can explain why the hedge ratio must be frozen before the trading period starts.

Step 2: Build the cointegration tools (K6)

Prompt - copy into your AI coding assistant
Create labs_fullcode/pipeline/pairs.py with numpy-only statistics: ols(y, X) returning coefficients, standard errors and residuals; adf_t(series, lags=1) - the ADF t-statistic of g in d e_t = g e_(t-1) + phi d e_(t-1); engle_granger(log_a, log_b) - step 1 OLS log A = a + beta log B, step 2 adf_t on the residual - returning (alpha, beta, adf_t); half_life(spread) from an AR(1) fit of spread changes; EG_CRIT = {'1%': -3.90, '5%': -3.34, '10%': -3.04} (MacKinnon, two variables with constant). FORMATION = ('2018-01-01', '2022-12-31'), TRADING = ('2023-01-01', '2025-12-31'). Rules: run from the labs folder; reuse labs_fullcode/common (config, data, broker, indicators, metrics, backtest, plotting) and earlier pipeline stages instead of rewriting them; do not modify common/ or data/; read every setting from pipeline/config.yaml via pipeline.config.load_config(); never put API keys or secrets in code; paper trading only; keep a main(argv) function so `python -m pipeline <command>` works; add short comments a finance professional can follow.

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Check: engle_granger on log KO and log PEP over the formation period returns beta about 0.756 and ADF t about -3.36.

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Step 3: Build the spread trading rules and leg targets (A4)

Prompt - copy into your AI coding assistant
In labs_fullcode/pipeline/pairs.py add trade_pair(closes, a, b, alpha, beta, entry, exit_, window, cost_bps): spread = log A - beta log B - alpha; z = (spread - rolling mean) / rolling std over window; state machine: short spread (-1: short A, long B) when z > +entry, long spread (+1) when z < -entry, exit when |z| < exit_; held = state.shift(1) (decided at close t, held over t+1); gross = held x w x (rA - beta rB) with w = 1 / (1 + |beta|); costs = |change in held| x w x (1 + |beta|) x cost_bps per leg per side; return spread, z, held, state, gross, net. leg_targets(state, a, b, beta) -> {a: state x w, b: -state x beta x w}. run(pair=('KO', 'PEP'), entry=2.0, exit_=0.5, window=60, cost_bps=5.0) for `python -m pipeline pairs --pair --entry --exit --window --cost-bps`: screen config universe.pairs (correlation of returns, beta, ADF t, cointegrated at 5%?), print the chosen pair's hedge ratio and half-life (warn if not cointegrated), trade 2023-2025 using 120 days of history for the z warm-up, print gross, net and SPY return, CAGR, volatility, Sharpe and max drawdown, trades, time in market, beta and correlation to SPY, the latest leg targets, and a note that short legs need a margin account while the automation stages are long only. Save chart Lab14_pairs_trading.png (normalised prices, z with bands and open positions, equity vs SPY). Rules: run from the labs folder; reuse labs_fullcode/common (config, data, broker, indicators, metrics, backtest, plotting) and earlier pipeline stages instead of rewriting them; do not modify common/ or data/; read every setting from pipeline/config.yaml via pipeline.config.load_config(); never put API keys or secrets in code; paper trading only; keep a main(argv) function so `python -m pipeline <command>` works; add short comments a finance professional can follow.

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Check: python -m pipeline pairs --help shows --pair, --entry, --exit, --window and --cost-bps.

Step 4: Run the pair screen (A4, K6)

Do this (no prompt needed)
Run: python -m pipeline pairs (same as: python3 Lab14-pairs-trading/pairs_stage.py). Compare correlation with the ADF t-statistic for each pair.

Check: KO/PEP: correlation 0.75, beta 0.76, ADF t -3.36 - cointegrated at 5% (critical -3.34). JPM/BAC 0.93, ADF -2.08 and XLF/JPM 0.94, ADF -1.89 - highly correlated but NOT cointegrated.

Step 5: Interpret the hedge ratio (K6)

Do this (no prompt needed)
Read section [2]. Explain in one sentence why the hedge ratio, not a 1:1 position, makes the spread stationary.

Check: KO/PEP hedge ratio 0.756 (short 0.76 of PEP per 1 of KO), half-life 39 days.

Step 6: Evaluate the out-of-sample result (A4)

Do this (no prompt needed)
Read section [3] and [4]: trades, time in market, beta to SPY and the latest leg targets.

Check: 2023-2025: 13 trades, 59% time in market, gross -9.4%, net -10.6% (Sharpe -0.60), beta to SPY +0.02, correlation +0.04, versus SPY +86.3%; latest z +0.65 -> spread position 0 -> leg targets KO 0.0, PEP 0.0. The hedge worked (beta near 0), but the spread did not revert profitably after 2022.

Step 7: Test a failed pair and other thresholds (A4, K6)

Do this (no prompt needed)
Run: python -m pipeline pairs --pair JPM BAC --entry 1.5 --exit 0. Explain why trading a pair that failed the cointegration test is a statistical error and why trying many pairs and thresholds is data snooping.

Check: JPM/BAC: 'WARNING: NOT cointegrated' (ADF -2.08, half-life 93 days); 1 trade held 100% of the time, net -21.9%, beta to SPY -0.03.

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Step 8: Study the chart (A4)

Prompt - copy into your AI coding assistant
Looking at labs_fullcode/_outputs/Lab14_pairs_trading.png and the stage output, explain to the head of trading how the desk should re-test cointegration on a rolling basis and size each pair small before this sleeve could ever be traded, and why it stays a paper sleeve while the automation stages are long only.

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Check: A rolling re-test plan (for example every quarter on the last 3 years) and a position-size limit per pair.

Run it

Step	Command / file
Run Lab 14	python3 Lab14-pairs-trading/pairs_stage.py (same as: python -m pipeline pairs)
The complete code	pipeline/pairs.py (reference copy: Lab14-pairs-trading/solution/)
Outputs	_outputs/ (charts, CSV files and reports written by the stage)

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